

Aryan Jalota

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EDUCATION

The University of Texas at Austin

Bachelor of Science in Computer Science & Mathematics, Minor in Statistics

Austin, TX

Aug 2024 – May 2028

EXPERIENCE

Salesforce

May 2026 – Aug 2026

Applied AI Software Engineering Intern (Office of the CEO)

San Francisco, CA

- Architected and shipped a distributed AI platform with a shared **Node.js/TypeScript** backend, **40+ REST APIs**, and an **MCP tool server** supporting production enterprise workflows.
- Improved AI query accuracy from **21%** to **95%** and reduced p50 latency **5×** through deterministic entity resolution, LLM routing, and multi-layer caching across live enterprise data.
- Reduced end-to-end information retrieval and preparation time from **2 hours** to **10 minutes (92%)** by building production retrieval and automation workflows across enterprise systems.
- Migrated a production backend from **SQLite** to **PostgreSQL** across **20+** tables and implemented **OAuth 2.0/Google SSO**, improving scalability and deployment reliability.

Depository Trust & Clearing Corporation

June 2025 – Aug 2025

Software Engineering Intern

Coppell, TX

- Engineered scalable **Python** pipelines to extract and validate multi-page audit reports, processing **1,000+** pages/hour with structured logging, fault tolerance, and automated validation, reducing manual workload by **10–15 hours/week**.

The University of Texas at Austin

Aug 2025 – May 2026

Undergraduate Research Software Engineer

Austin, TX

- Designed a distributed data pipeline using parallel and asynchronous processing to transform geospatial datasets into optimized **HDF5** feature stores, improving ML training throughput by **45%**.
- Built modular end-to-end ML pipelines in **Python** using **PyTorch** and **NumPy**, with **PyTest**, CI workflows, and automated data validation, improving reliability and reproducibility across data-intensive research systems.

The University of Texas at Austin

Aug 2024 – May 2026

Undergraduate Research Software Engineer

Austin, TX

- Designed and optimized real-time computer vision systems using **PyTorch**, **YOLOv8**, and **OpenCV**, leveraging **GPU-accelerated** training and inference for low-latency vehicle detection.
- Validated deep learning models through systematic training and validation workflows, achieving **0.85+ mAP** with **high-80s** precision/recall for deployment-oriented traffic analysis.

PROJECTS

Distributed Lakehouse Analytics Platform | Python, Apache Spark, Delta Lake, FastAPI, Redis

- Architected a distributed lakehouse on **Apache Spark** and **Delta Lake** supporting **ACID** transactions and **SQL** analytics across **500M+** records.
- Reduced repeated-query latency **60%** with a **FastAPI** query service using **Redis** caching and **async** job scheduling.

Distributed Search Engine | C++, Go, gRPC, Redis, RocksDB, Docker

- Built a distributed search engine indexing **10M+ documents** with **inverted indexes** and **BM25** ranking at sub-100 ms query latency across shards.
- Designed a distributed **gRPC** query engine with consistent hashing and **Redis** caching, reducing query latency **65%**; parallelized **RocksDB** indexing with a multithreaded C++ pipeline to achieve **4×** higher throughput.

High-Performance Trading Engine | C++17, Multithreading, Linux, CMake, Google Benchmark

- Engineered a high-performance **C++17** limit order book processing **1M+** market events/sec using cache-efficient data structures and custom memory management, validated through **Google Benchmark**.
- Parallelized order processing with **std::thread** and producer-consumer queues for **3×** throughput.

TECHNICAL SKILLS

Languages: C++17, Python, Go, Java, TypeScript, JavaScript, SQL **GPU/ML:** CUDA, PyTorch, GPU Computing, Deep Learning, Computer Vision, OpenCV, YOLOv8, NumPy **Systems:** Operating Systems, Distributed Systems, Concurrency, Parallel Computing, Systems Programming, Performance Optimization, Data Structures & Algorithms, Debugging, Object-Oriented Programming **Infrastructure:** Linux, Docker, Kubernetes, Microservices, Containers, gRPC, REST APIs, Git, CI/CD, AWS **Data & Storage:** Apache Spark, Delta Lake, PostgreSQL, MySQL, SQLite, Redis, RocksDB, HDF5 **Frameworks:** Node.js, Express, FastAPI, React, Spring Boot